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Interview Q&A: deep learning

1. What is the main difference between supervised and unsupervised learning?

Supervised learning involves training a model on labeled data to make predictions, whereas unsupervised learning involves training a model on unlabeled data to identify patterns or relationships.

2. What is the purpose of activation functions in deep learning?

Activation functions introduce non-linearity into neural networks, allowing them to learn complex relationships between inputs and outputs.

3. What is the primary goal of pre-training in deep learning?

Pre-training aims to learn general representations that can be fine-tuned for specific tasks, reducing the need for large amounts of task-specific data.

4. What is the concept of overfitting in deep learning?

Overfitting occurs when a model is too complex and learns the noise in the training data, resulting in poor performance on new, unseen data.

5. What is the difference between a feedforward neural network and a recurrent neural network?

A feedforward neural network is a type of neural network where data flows only in one direction, from input layer to output layer without any feedback loops. A recurrent neural network (RNN) is a type of neural network that uses feedback connections to allow information to persist over time.

6. What is the role of batch normalization in deep learning?

Batch normalization is a technique used to normalize the input data for each layer, reducing internal covariate shift and improving training stability.

7. What is the concept of gradient descent in deep learning?

Gradient descent is an optimization algorithm used to minimize the loss function by iteratively adjusting the model's parameters.

8. What is the difference between a convolutional neural network (CNN) and a recurrent neural network (RNN)?

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A CNN is designed for image and signal processing tasks, using convolutional and pooling layers to extract spatial features, whereas an RNN is suited for sequential data such as text or time series data, utilizing recurrent connections to capture temporal dependencies.

9. What is the difference between a deep neural network and a shallow neural network?

A deep neural network has multiple hidden layers, whereas a shallow neural network typically only has one or two hidden layers.

10. What is the concept of backpropagation in deep learning?

Backpropagation is an algorithm used to train artificial neural networks by minimizing the error between predicted outputs and actual outputs.

11. What is the purpose of using convolutional neural networks (CNNs) in image classification tasks?

CNNs are designed to efficiently process data with grid-like topology, such as images, by applying small filters over the entire input.

12. What is the difference between a generative model and a discriminative model in deep learning?

Generative models, such as GANs and VAEs, learn to generate new data samples that resemble the training data, whereas discriminative models, such as logistic regression and neural networks, learn to classify or predict labels for given input data.

13. What is the role of regularization techniques in deep learning?

Regularization techniques, such as dropout and L1/L2 regularization, are used to prevent overfitting by adding a penalty term to the loss function that discourages large weights.

14. What is the purpose of using dropout regularization in deep learning?

Dropout regularization randomly drops out neurons during training to prevent overfitting by reducing the model's reliance on any single neuron.

15. What is the difference between a one-hot encoding and label smoothing?

One-hot encoding assigns a probability of 1 to the correct class and 0 to all other classes, while label smoothing reduces this to a small value (e.g., 0.9 for the correct class and 0.05 for all other classes) to prevent overconfidence in the model's predictions.